Ar & Mr

Exam code: 0620 & 0971

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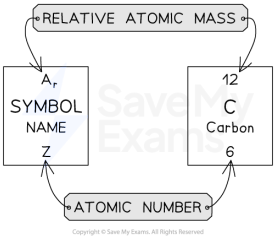
Relative masses

Relative atomic mass

- The symbol for relative atomic mass is A_r
- The relative atomic mass for each element can be found in the Periodic Table, along with the atomic number
  - Relative atomic mass is shown on the atomic symbol
  - It is always larger than the atomic number (except for hydrogen, where they are the same)
  - Use the key on the Periodic Table to correctly identify the mass number

Help me understand

Key for chemical symbols on the Periodic Table



This key is given on the Periodic Table in exams and identifies the number that is the relative atomic mass.

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Got it

fixed mass of 12 units
    - The mass of all other atoms are compared against carbon-12

- The relative atomic mass of carbon is 12
    - The relative atomic mass of magnesium is 24 which means that magnesium is twice as heavy as carbon
    - The relative atomic mass of hydrogen is 1 which means it has one-twelfth the mass of one carbon-12 atom

Relative molecular (formula) mass

- The symbol for the relative molecular mass is M_r
- Relative molecular mass is the sum of the relative atomic masses of all the atoms in a molecule
    - The term **relative formula mass** is used when referring to the total mass of an **ionic** compound

- To calculate the M_r of a substance, you have to add up the relative atomic masses of all the atoms present in the formula

Relative Formula Mass Calculations Table

Substance	Atoms present	Calculation	M_r
Hydrogen H_2	2 x H	(2 x 1)	2
Water H_2O	(2 x H) + (1 x O)	(2 x 1) + (1 x 16)	18
Potassium carbonate K_2CO_3	(2 x K) + (1 x C) + (3 x O)	(2 x 39) + (1 x 12) + (3 x 16)	138
Calcium hydroxide $Ca(OH)_2$	(1 x Ca) + (2 x O) + (2 x H)	(1 x 40) + (2 x 16) + (2 x 1)	74
Ammonium sulfate $(NH_4)_2SO_4$	(2 x N) + (8 x H) + (1 x S) + (4 x O)	(2 x 14) + (8 x 1) + (1 x 32) + (4 x 16)	132

Worked Example

Calculate the relative formula mass of:

1. Sodium chloride, NaCl
2. Copper oxide, CuO
3. Magnesium nitrate, $Mg(NO_3)_2$

Answers:

1. Sodium chloride

3. Magnesium nitrate
    - $Mg(NO_3)_2 = 24 + (14 \times 1 \times 2) + (16 \times 3 \times 2) = 148$

Reacting masses

- The Law of Conservation of mass tells us that mass cannot be created or destroyed
    - In a chemical reaction, the total mass of reactants equals the total mass of the products
- We can use this, along with relative atomic / formula masses to perform calculations to identify the quantities of reactants or products involved in a chemical reaction
- **Example**:

$$2Ca + O_2 \rightarrow 2CaO$$

- Relative atomic masses: Ca = 40; O = 16
- Using the balanced symbol equation:
    - Reactants:
    - $2 \times 40 = 80$ units of mass of calcium
    - $2 \times 16 = 32$ units of mass of oxygen (O_2 molecule, 16 + 16 = 32)
    - Products:
    - $2 \times (40 + 16) = 112$ units of mass of CaO

$$2Ca + O_2 \rightarrow 2CaO$$

$$80 + 32 = 112$$

- The ratio of the mass of calcium and oxygen reacting will always be the same, regardless of the units
    - 80 g of calcium will react with 32 g of oxygen to form 112 g of calcium oxide
    - 80 tonnes of calcium will react with 32 tonnes of oxygen to form 112 tonnes of calcium oxide
    - So, 40 kg of calcium will react with 16 kg of oxygen to form 56 kg of calcium oxide

Worked Example

Calculate the mass of carbon dioxide produced when 32 g of methane, CH_4, reacts completely in excess oxygen:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

Relative atomic masses, A_r: H = 1; C = 12; O = 16

Answer:

- Using the balanced symbol equation:
    - Reactants:
    - $12 + (4 \times 1) = 16$ units of mass of methane
    - $2 \times 16 = 32$ units of mass of oxygen (O_2 molecule, 16 + 16 = 32)
    - Products:
    - $12 + (2 \times 16) = 44$ units of mass of carbon dioxide

$$16 + 64 \rightarrow 44 + 36$$

- So, 16 g of methane would react in excess oxygen to form 44 g of carbon dioxide
- 32 g of methane is double the amount of methane from the balanced symbol equation
- So, this would produce double the amount of caron dioxide
  - 2 x 44 = **88 g** of carbon dioxide

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